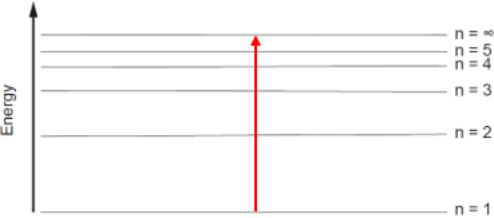
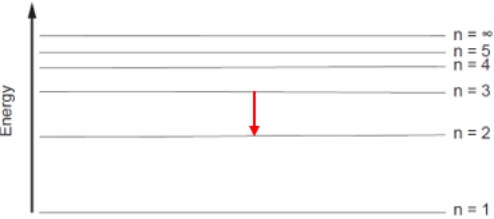


SECTION A

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1.				linear (1) 3 bonding pairs and 1 lone pair award (1) for both	2			2		
2.				metallic (1) covalent and van der Waals award (1) for both	2			2		
3.				+6 / +VI do not accept 6+		1		1		
4.	(a)				1			1		
	(b)				1			1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5.				agree both have 18 electrons / same number of electrons / same number of electrons as an argon atom / electronic configuration of $1s^2 2s^2 2p^6 3s^2 3p^6$			1	1		
6.				$0.78 = \frac{4.2}{\text{mass of reactants}}$ (1) mass of reactants = $\frac{4.2}{0.78} = 5.4 \text{ g}$ (1) alternative method $n(\text{Cu}) \text{ formed} = \frac{4.2}{63.5} = 0.066 \text{ mol}$ (1) $0.066 \text{ mol of CuO} + 0.066 \text{ mol of H}_2 = 5.25 + 0.13 = 5.38 \text{ g}$		2		2	1	
				Section A total	6	3	1	10	1	0